

Product Requirement Document (PRD)

Aplikasi Pinjaman Online - PT. XYZ

Mochamad Fadlan Abdillah

Versi 1.0 | Tanggal 2026-06-15

1. Purpose

Dokumen ini berisi persyaratan fungsional untuk aplikasi pinjaman online mobile PT. XYZ. Lingkup dokumen ini adalah tahap design specification sebagai materi technical assessment posisi Solution Analyst.

MVP harus mendukung flow bisnis berikut:

- *Registrasi pengguna dilengkapi dengan data diri, email, nomor telepon dan upload foto selfie sambil memegang KTP.*
- *Login menggunakan password atau biometrik perangkat (jika tersedia).*
- *Tampilan sisa hutang dan tagihan bulanan yang harus dibayarkan.*
- *Pengajuan pinjaman maksimum Rp 12.000.000 dengan tenor maksimal 12 bulan.*
- *Proses decision engine: hasil approved atau rejected.*
- *Notifikasi dikirim melalui email dan SMS setelah pinjaman disetujui.*
- *Blokir pengajuan baru selama ada pinjaman berstatus pending atau active.*

2. Scope

2.1 In-Scope

- *Registrasi dan proses KYC (upload KTP + selfie) menggunakan signed URL ke object storage.*
- *Autentikasi user (email/phone + password) ditambah opsi biometric lokal untuk unlock token.*
- *Pengajuan pinjaman maksimal Rp 12.000.000 dengan tenor paling lama 12 bulan.*
- *Mesin pengambilan keputusan otomatis (decision engine) berbasis aturan yang menghasilkan hasil konsisten (rule-based dan deterministik).*
- *Pembuatan jadwal cicilan bulanan setelah pinjaman disetujui.*
- *Endpoint pencatatan pembayaran dengan webhook settlement pada sandbox.*
- *Notifikasi melalui Email dan SMS.*

2.2 Out-of-Scope

- *Integrasi ke lembaga penilaian kredit sungguhan (credit bureau).*
- *Pencairan dana langsung ke rekening nasabah (direct disbursement).*
- *Rekonsiliasi perbankan.*
- *Push notification.*

3. Success Criteria

- *Specification Quality: Spesifikasi jelas tanpa ambiguitas sehingga developer dapat langsung memulai implementasi MVP.*
- *E2E Functional: End-to-end flow berjalan dari registrasi sampai pembayaran.*
- *Deterministic: Hasil decision engine sama setiap kali dengan input identik.*
- *Achievable: Tim kecil (1-2 developer) dapat build MVP dalam 3-5 minggu.*

4. Environment Assumptions

-
- *Verifikasi KYC otomatis dan deterministik (tidak ada review manual di environment test).*
 - *Email ditampung oleh local SMTP mock.*
 - *SMS via stub HTTP sederhana.*
 - *Pembayaran via payment sandbox dengan webhook mock.*
 - *Tech stack dapat disesuaikan sesuai preferensi tim.*

5. Deliverables

-
- *PRD: Product Requirements Document (dokumen ini).*
 - *FSD: Functional Specification Document (dokumen terpisah).*
 - *ERD: diagram ER terdapat di FSD.*
 - *API Spec: OpenAPI YAML (bisa import ke Swagger UI / FastAPI).*

Functional Specification Document(FSD)

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1. High Level Architecture

Sistem terbagi menjadi empat layer utama dengan komunikasi berlapis:

- Mobile Client Layer: React Native, single codebase untuk Android dan iOS. Token autentikasi disimpan di Keychain (iOS) / Keystore (Android). Biometric hanya sebagai local unlock untuk mengakses token.
- Backend API Layer: REST/JSON API dengan JWT (access token 15 menit, refresh token 30 hari). Kontrak mengikuti OpenAPI 3.0.
- Data Storage Layer: PostgreSQL untuk relasi entitas inti. MinIO (S3-compatible) untuk penyimpanan file KTP dan selfie.
- Background Worker Layer: Redis untuk job queue: verifikasi KYC, decision engine, notifikasi. Masing-masing worker independen.
- External Service Simulation: SMTP mock, stub HTTP (SMS), payment sandbox (webhook mock).

1.1 Component Flow

Mobile App -> API Gateway -> Auth Service -> User/KYC Service -> Loan Service -> Payment Service -> Notification Service -> PostgreSQL

Loan Service -> Redis Queue -> KYC Verification Worker -> Loan Decision Worker -> Notification Worker -> PostgreSQL/MinIO/Email-SMS

Payment Service -> Payment Sandbox -> Webhook -> Payment Worker -> update database

1.2 Technology Stack Recommendation

Bagian Sistem	Teknologi Terpilih	Alasan Pemilihan
Mobile App	React Native (Expo)	Satu basis kode untuk Android dan iOS;
Server API Backend	FastAPI (Python)	Dokumentasi Swagger/OpenAPI otomatis; async native
Database Relasional	PostgreSQL	Transaksional ACID; dukungan query kompleks
Penyimpanan File	MinIO (lokal) / S3	Terpisah dari database; skalabel untuk file
Job Queue	Redis + RQ	Ringkas; stabil; ekosistem luas
Authentication	JWT (akses + refresh)	Tanpa state di server; pola industri standar

2. Screen Flow

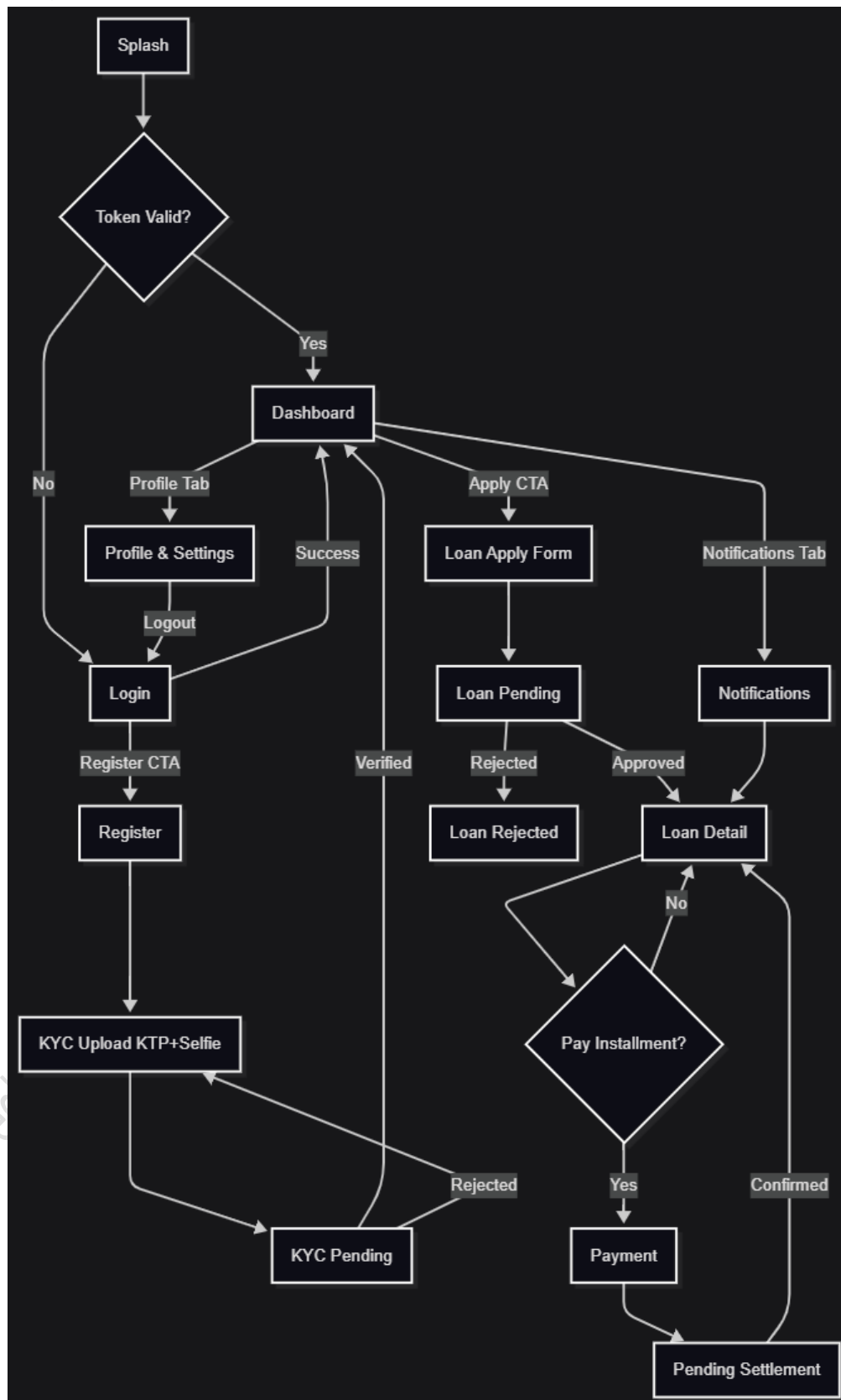
2.1 Daftar Navigasi Layar

Nama Layar	Route	Tujuan
Splash	/splash	Cek token, tentukan navigasi awal
Login	/login	Verifikasi identitas user

Register	/register	Pembuatan akun baru
KYC Upload	/kyc/upload	Upload KTP, selfie
KYC Status	/kyc/pending	Tampilkan progress verifikasi
Dashboard	/dashboard	Ringkasan hutang dan tagihan
Loan Apply	/loan/apply	Input jumlah dan tenor pinjaman
Loan Pending	/loan/pending	Menunggu hasil keputusan
Loan Detail	/loan/{id}	Info pinjaman dan jadwal cicilan
Payment	/loan/{id}/pay	Form pembayaran cicilan
Notifications	/notifications	Inbox pesan sistem
Profile & Settings	/profile	Data pribadi, toggle biometric, logout

Technical Assessment - Solution Analyst - Mochamad Fadlan Abdillahi

2.2 Navigation Flow



3. Entity Relationship Diagram (ERD)

3.1 Tabel Entitas Inti

Nama Tabel	Peran
USER	Data pribadi pengguna yang login ke aplikasi mobile
IDENTITY_DOCUMENT	File bukti diri yang diunggah pengguna
LOAN	Setiap permintaan pinjaman dari pengguna
REPAYMENT_SCHEDULE	Jadwal cicilan bulanan untuk setiap pinjaman
PAYMENT	Catatan setiap transaksi pembayaran

3.2 ERD

USER

Column	Type	Constraint
<u>id</u>	UUID	PK
<u>full_name</u>	VARCHAR	NOT NULL
<u>email</u>	VARCHAR	UNIQUE, NOT NULL
<u>phone</u>	VARCHAR	UNIQUE, NOT NULL
<u>password_hash</u>	VARCHAR	NOT NULL
<u>kyc_status</u>	VARCHAR	NOT NULL
<u>is_biometric_enabled</u>	BOOLEAN	DEFAULT FALSE
<u>created_at</u>	TIMESTAMP	NOT NULL
<u>updated_at</u>	TIMESTAMP	NOT NULL

IDENTITY_DOCUMENT

Column	Type	Constraint
<u>id</u>	UUID	PK
<u>user_id</u>	UUID	FK → USER.id
<u>document_type</u>	VARCHAR	NOT NULL
<u>file_url</u>	TEXT	NOT NULL
<u>verification_status</u>	VARCHAR	NOT NULL
<u>uploaded_at</u>	TIMESTAMP	NOT NULL
<u>verified_at</u>	TIMESTAMP	NULLABLE
<u>created_at</u>	TIMESTAMP	NOT NULL

LOAN

Column	Type	Constraint
<u>id</u>	UUID	PK
<u>user_id</u>	UUID	FK → USER.id
<u>amount_requested</u>	DECIMAL(18,2)	NOT NULL
<u>amount_approved</u>	DECIMAL(18,2)	NULLABLE

<i>tenor_months</i>	INTEGER	NOT NULL
<i>interest_rate</i>	DECIMAL(5,2)	NOT NULL
<i>processing_fee</i>	DECIMAL(18,2)	DEFAULT 0
<i>admin_fee</i>	DECIMAL(18,2)	DEFAULT 0
<i>late_fee_daily</i>	DECIMAL(18,2)	DEFAULT 0
<i>loan_status</i>	VARCHAR	NOT NULL
<i>applied_at</i>	TIMESTAMP	NOT NULL
<i>approved_at</i>	TIMESTAMP	NULLABLE
<i>disbursed_at</i>	TIMESTAMP	NULLABLE
<i>rejected_at</i>	TIMESTAMP	NULLABLE
<i>created_at</i>	TIMESTAMP	NOT NULL
<i>updated_at</i>	TIMESTAMP	NOT NULL

REPAYMENT_SCHEDULE

Column	Type	Constraint
<i>id</i>	UUID	PK
<i>loan_id</i>	UUID	FK → LOAN.id
<i>installment_number</i>	INTEGER	NOT NULL
<i>due_date</i>	DATE	NOT NULL
<i>principal_amount</i>	DECIMAL(18,2)	NOT NULL
<i>interest_amount</i>	DECIMAL(18,2)	NOT NULL
<i>penalty_amount</i>	DECIMAL(18,2)	DEFAULT 0
<i>total_amount</i>	DECIMAL(18,2)	NOT NULL
<i>paid_amount</i>	DECIMAL(18,2)	DEFAULT 0
<i>repayment_status</i>	VARCHAR	NOT NULL
<i>paid_at</i>	TIMESTAMP	NULLABLE
<i>created_at</i>	TIMESTAMP	NOT NULL
<i>updated_at</i>	TIMESTAMP	NOT NULL

PAYMENT

Column	Type	Constraint
<i>id</i>	UUID	PK
<i>loan_id</i>	UUID	FK → LOAN.id
<i>repayment_schedule_id</i>	UUID	FK → REPAYMENT_SCHEDULE.id
<i>amount</i>	DECIMAL(18,2)	NOT NULL
<i>payment_method</i>	VARCHAR	NOT NULL
<i>payment_channel</i>	VARCHAR	NOT NULL
<i>provider_reference</i>	VARCHAR	UNIQUE
<i>payment_status</i>	VARCHAR	NOT NULL
<i>paid_at</i>	TIMESTAMP	NULLABLE
<i>created_at</i>	TIMESTAMP	NOT NULL

Relationships

Parent Table	Relationship	Child Table
USER	1 → N	IDENTITY_DOCUMENT
USER	1 → N	LOAN

LOAN	1 → N	REPAYMENT_SCHEDULE
LOAN	1 → N	PAYMENT
REPAYMENT_SCHEDULE	1 → N	PAYMENT

4. API Contract (OpenAPI)

4.1 Uniform Error Model

```
{
  "error": {
    "code": "INVALID_INPUT",
    "message": "Gagal validasi",
    "details": [{ "field": "email", "message": "Format email salah" }]
  }
}
```

4.2 Public Endpoints

Metode	Endpoint	Isi Body	Hasil Sukses
POST	/auth/register	{"full_name", "email", "phone", "password"}	201 Created, kyc_status = pending
POST	/auth/login	{"identifier" (email/phone), "password"}	200 OK, token JWT

4.3 Authenticated Endpoints (Bearer Auth)

Aksi	Endpoint	Body / Parameter	Keterangan Hasil
Upload URL	POST /files/upload-url	{"file_name", "content_type", "purpose"}	200: upload_url, file_id, file_url
Submit KYC	POST /user/kyc	{"document_ids[]", "selfie_file_id"}	202: kyc_status = pending
Profil Saya	GET /user/me		200: user + ringkasan pinjaman
Ajukan Pinjaman	POST /loans	{"amount", "tenor_months", "idempotency_key"}	202: pending / 409: conflict / 422: unprocessable
Daftar Pinjaman	GET /loans		200: array pinjaman
Detail Pinjaman	GET /loans/:id		200: object pinjaman
Jadwal Cicilan	GET /loans/:id/schedule		200: array cicilan per bulan
Bayar	POST /payments	{"loan_id", "amount", "method", "provider_reference"}	201: payment_id, status pending
Notifikasi	GET /notifications		200: array notifikasi
Override KYC (test)	POST /internal/test/kyc/:user_id	{"status": "verified/rejected"}	200: override (khusus admin test)
Override Keputusan (test)	POST /internal/test/loans/:id/decision	{"status", "reason_code"}	200: override (khusus admin test)

4.4 Request-Response Examples

POST /auth/register

Request Body:

```
{
  "full_name": "Anton Wijaya",
  "email": "anton@mail.com",
  "phone": "+628123456789",
  "password": "P@ssw0rd"
}
```

Response 201 Created:

```
{
  "id": "uuid-xxxxxxx",
  "kyc_status": "pending"
}
```

POST /loans (Ajukan Pinjaman)

Request Body:

```
{
  "amount": 5000000,
  "tenor_months": 6,
  "purpose": "Modal usaha",
  "idempotency_key": "uuid-yyyyyyyy"
}
```

Response 202 Accepted (masuk antrean keputusan):

```
{
  "loan_id": "loan_abc123",
  "status": "pending",
  "estimated_time_seconds": 30
}
```

Response 409 Conflict (ada pinjaman aktif/pending):

```
{
  "error": {
    "code": "CONFLICT_ACTIVE_LOAN",
    "message": "Anda masih memiliki pinjaman yang sedang berlangsung."
  }
}
```

Response 422 Unprocessable Entity (KYC belum diverifikasi):

```
{
  "error": {
    "code": "KYC_NOT_VERIFIED",
    "message": "Harap selesaikan verifikasi KTP terlebih dahulu."
  }
}
```

GET /loans/{id}/schedule (Jadwal Cicilan)

Response 200 OK:

```
[
  {
    "id": "r_001",
    "due_date": "2026-06-12",
    "principal_due": 781250,
    "interest_due": 62500,
    "total_due": 843750,
    "status": "due"
  }
]
```

```

},
{
  "id": "r_002",
  "due_date": "2026-07-12",
  "principal_due": 787500,
  "interest_due": 56250,
  "total_due": 843750,
  "status": "due"
}
]

```

POST /payments (Pencatatan Pembayaran)

Request Body:

```

{
  "loan_id": "loan_abc123",
  "amount": 843750,
  "method": "VA",
  "provider_reference": "INV-20260511-001"
}

```

Response 201 Created:

```

{
  "id": "pay_xyz789",
  "loan_id": "loan_abc123",
  "amount": 843750,
  "status": "pending",
  "created_at": "2026-05-11T14:30:00Z"
}

```

4.5 Security Notes

- Transport: *Wajib HTTPS/TLS untuk semua endpoint.*
- Password Hashing: *Password di-hash dengan bcrypt (work factor minimal 12).*
- JWT Token Management: *Access token 15 menit, refresh token 30 hari (single use).*
- Signed Upload URLs: *File KTP/selfie hanya via signed URL (expiry default 5 menit).*
- Rate Limiting: *Max 5 failed login attempts dalam 10 menit, lalu lock sementara.*

5. Loan Decision Engine (Rule-Based, Deterministic)

5.1 Worker Pipeline

Loan application via POST /loans -> Redis queue.

Worker:

1. *cek KYC status,*
2. *hitung scoring,*
3. *putusan approved/rejected,*
4. *update loan record,*
5. *buat repayment schedule,*
6. *irim notifikasi email & SMS.*

5.2 Hard Reject Rules

Reject Condition	Reason Code
<i>KYC status != verified</i>	<i>KYC_NOT_VERIFIED</i>
<i>Existing loan with status pending OR active</i>	<i>CONFLICT_ACTIVE_LOAN</i>

5.3 Scoring

Variable	Condition	Score
<i>phone_verified</i>	<i>true</i>	<i>+10</i>
<i>prior_repayments_count</i>	<i>0</i>	<i>0</i>
<i>prior_repayments_count</i>	<i>1-2</i>	<i>+10</i>
<i>prior_repayments_count</i>	<i>>= 3</i>	<i>+20</i>
<i>device_age_days</i>	<i>> 180</i>	<i>+10</i>
<i>request_amount</i>	<i><= Rp 2.000.000</i>	<i>+5</i>

5.4 Decision Threshold

Total Score	Decision	Approved Amount
<i>>= 30</i>	<i>APPROVED_FULL</i>	<i>Full requested amount</i>
<i>20 - 29</i>	<i>APPROVED_PARTIAL</i>	<i>floor(request_amount * 2/3)</i>
<i>< 20</i>	<i>REJECTED</i>	<i>Rp 0</i>

5.5 Test Vectors

Case	KYC Status	Verified?	Prior Repay	Device Age	Amount	Result	Score
A	<i>Verified</i>	<i>Yes</i>	<i>5</i>	<i>300 days</i>	<i>Rp 5.000.000</i>	<i>Approved Full</i>	<i>40</i>
B	<i>Verified</i>	<i>No</i>	<i>0</i>	<i>10 days</i>	<i>Rp 10.000.000</i>	<i>Rejected (low score)</i>	<i>0</i>
C	<i>Verified</i>	<i>Yes</i>	<i>2</i>	<i>10 days</i>	<i>Rp 1.000.000</i>	<i>Approved Partial</i>	<i>25</i>
D	<i>Pending</i>	<i>Yes</i>	<i>3</i>	<i>400 days</i>	<i>Rp 3.000.000</i>	<i>Rejected (KYC)</i>	<i>auto reject</i>
E	<i>Verified</i>	<i>Yes</i>	<i>0</i>	<i>50 days</i>	<i>Rp 12.000.000</i>	<i>Approved Partial</i>	<i>20</i>

6. Screen Behavior Details

6.1 Splash Screen

- *Tujuan: Validasi access token dan refresh token yang tersimpan secara lokal sebelum menampilkan konten aplikasi.*
- *Initial State: Baca storage lokal (Keychain/Keystore) untuk access_token dan refresh_token.*
- *Transisi:*
 - *(a) access_token valid -> langsung ke Dashboard.*

- o (b) Hanya ada refresh_token -> POST /auth/refresh -> sukses ke Dashboard, gagal ke Login. (c) Tidak ada token -> arahkan ke Login.
- Error: Jika refresh token expired, hapus kedua token dari storage lalu redirect ke Login.
- Edge Case: Access token corrupt -> bersihkan session dan redirect ke Login.

6.2 Login Screen

- Input Fields: Identifier (email atau phone) dan password.
- Client Validation: Identifier wajib diisi (format email atau digit nomor ponsel). Password minimal 8 karakter (huruf besar, kecil, angka, simbol).
- Server Flow: POST /auth/login -> access_token + refresh_token. Simpan di Keychain/Keystore. Tawarkan enable biometric setelah login sukses.
- Local Biometric: Enkapsulasi refresh_token agar hanya bisa dibaca setelah verifikasi sidik jari/wajah. Biometric tidak mengubah autentikasi di server.
- Error Response: 401 -> 'Email atau password salah' (tidak menyebut field spesifik). 429 -> 'Terlalu banyak percobaan, coba lagi nanti.'
- Edge Case: Offline -> alert 'Gangguan jaringan' + tombol Retry. Tombol Login disabled saat loading.

6.3 Register Screen

- Input Fields: Nama Lengkap, Email, Phone (prefix +62), Password, Confirm Password.
- Validation: Semua field wajib. Format email sesuai RFC 5322. Phone berupa digit (prefix +62 opsional). Password minimal 8 karakter. Konfirmasi password HARUS sama. Server: 201 -> sukses lanjut ke KYC Upload. 409 -> email/phone sudah dipakai. 400 -> format salah, highlight field bermasalah.
- Interaction: Tombol Register disabled selama request berjalan (cegah double click).
- Edge Case: User tekan Back saat request berjalan -> cancel request dan kembali tanpa kehilangan data terisi.

6.4 KYC Upload Screen

- Komponen upload:
 - o KTP
 - o Selfie (dari kamera atau galeri).
- Flow per File:
 - o (a) Minta signed URL via POST /files/upload-url (filename, content_type).
 - o (b) PUT gambar ke signed URL.
 - o (c) Simpan file_id. Ulangi sampai ketiga file selesai, lalu POST /user/kyc dengan ketiga file_id.
- Image Pre-processing: Kompresi client-side ~1 MB. Koreksi orientasi dari EXIF. Tampilkan preview thumbnail untuk rotasi manual.
- State Machine: IDLE -> Uploading (file1/file2/file3) -> All Done -> Submit -> KYC Pending Screen.
- Retry: Maksimal 3 percobaan per file. Jika tetap gagal, arahkan ke customer service.

- *Edge Case: User cancel di tengah proses -> konfirmasi sebelum hapus file yang sudah terupload.*

6.5 KYC Pending Screen

- *Visual: Judul 'Processing' + spinner animasi + estimasi ~30 detik.*
- *Polling: GET /user/me setiap 5 detik, cek kyc_status.*
- *Transisi:*
 - o *(a) verified -> redirect ke Dashboard.*
 - o *(b) rejected -> kembali ke KYC Upload dengan alasan penolakan.*
- *Timeout: Jika polling > 2 menit -> 'Waktu proses melebihi estimasi' + tombol hubungi CS.*
- *Edge Case: User close app saat pending -> saat buka lagi, polling dimulai ulang.*

6.6 Dashboard Screen

- *Data Sources: GET /user/me (profile) + GET /loans (loan list).*
- *Layout: Header (nama user + notification bell badge). Main area:*
 - o *(a) Ada loan aktif -> ringkasan outstanding balance, due date, tagihan bulan ini.*
 - o *(b) Sedang pending -> card 'Loan Sedang Diproses' + link ke detail.*
 - o *(c) Belum ada loan -> 'Belum Ada Pinjaman' + CTA Ajukan. Tombol Ajukan disabled jika KYC belum verified atau ada loan pending/active.*
- *Refresh: Pull-to-refresh untuk update semua widget secara paralel.*
- *Error: 'Gagal Memuat Data' + tombol Retry.*
- *Edge Case: Navigasi ke Loan Detail dari notifikasi.*

6.7 Loan Apply Screen

- *Controls: Slider nominal (min Rp 100.000 - max Rp 12.000.000). Tenor picker (1-12 bulan). Field tujuan penggunaan (optional).*
- *Real-time Preview: Saat slider digerakkan, preview auto-update: bunga per bulan, biaya admin, total pembayaran, cicilan per bulan.*
- *Client Validation: Amount >= 100.000 dan <= 12.000.000. Tenor 1-12 bulan.*
- *Pre-submit Check:*
 - o *(a) KYC status == verified. Jika belum -> modal warning.*
 - o *(b) Tidak ada loan pending/active lain. Jika ada -> infokan user selesaikan kewajiban sebelumnya.*
- *Process: POST /loans dengan idempotency_key (generate UI random tiap submit). Response 202 -> ke Pending Screen. Response 409/422 -> tampilkan pesan sesuai error code.*
- *Edge Case: Network timeout -> jangan resend (cegah duplikasi). Polling GET /loans/{id} untuk cek status.*

6.8 Loan Pending Screen

- *Visual: Judul 'Processing Decision' + progress bar animasi (~30 detik).*
- *Polling: GET /loans/{id} tiap 3 detik (exponential backoff max 40 detik).*
- *Transisi: Approved -> Loan Detail. Rejected -> Loan Rejected Screen.*
- *Timeout: > 1 menit -> 'Proses memerlukan pemeriksaan tambahan' + kontak CS.*

- *Edge Case: App ke background -> polling stop. Masuk foreground -> polling lanjut.*

6.9 Loan Rejected Screen

- *Ikon: Silang merah atau warning icon.*
- *Judul: 'Pinjaman Ditolak'.*
- *Penjelasan: reason_code singkat (tanpa skor internal).*
- *Actions: Kembali ke Dashboard / Hubungi Customer Support.*

6.10 Loan Detail Screen

- *Data: GET /loans/{id} + GET /loans/{id}/schedule.*
- *Layout: Header card (badge status). Info pokok (nominal approved, plafon, tenor, bunga). Progress bar cicilan (lunas/total). Tabel angsuran (due date, nominal, status, merah jika overdue).*
- *Actions: Bayar Cicilan -> payment form. Download Schedule (next stage).*
- *Empty State: Jika baru approved tapi schedule belum siap -> 'Jadwal cicilan sedang disiapkan, mohon tunggu.'*

6.11 Payment Screen

- *Fields: Amount (auto-fill dari target cicilan), method (VA/Transfer/Retail).*
- *API: POST /payments -> 201 {payment_id, status: pending}.*
- *Display: Instruksi pembayaran (VA number, nominal, expiry time).*
- *Settlement: Webhook -> status settled -> navigasi ke Loan Detail + refresh.*
- *Error: 'Gagal membuat pembayaran, silakan ulangi.'*

6.12 Notification Inbox Screen

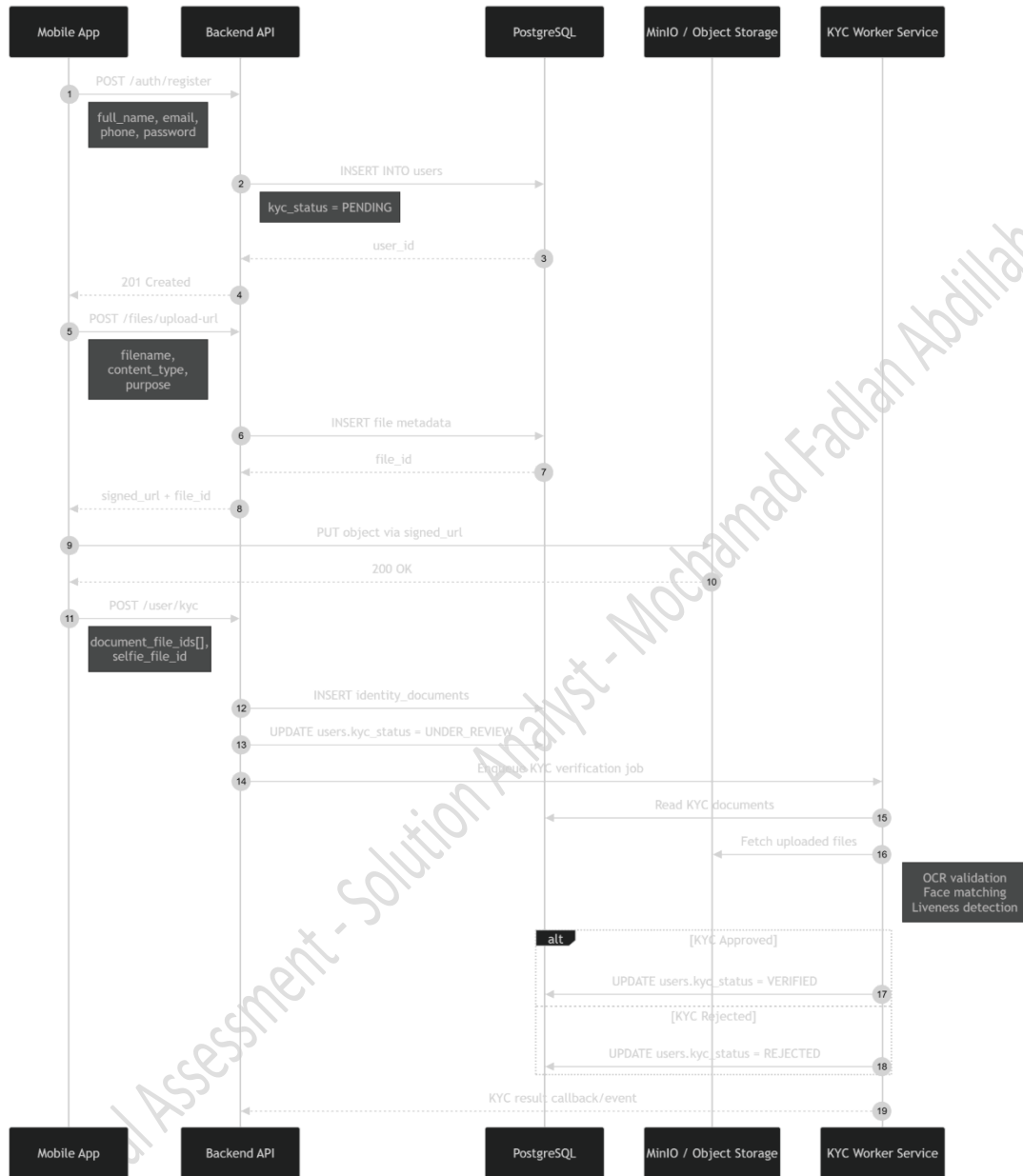
- *Data: GET /notifications.*
- *Display: Icon channel, title, preview body, timestamp, read/unread marker. Tap -> mark read=true, buka konteks terkait (loan detail, dll).*

6.13 Profile & Settings Screen

- *Akses: Profil lengkap + status verifikasi KYC.*
- *Toggle: Biometric autofill (hanya jika device mendukung).*
- *Logout: Hapus keychain, hapus token, redirect ke Login.*

7. Sequence Diagrams

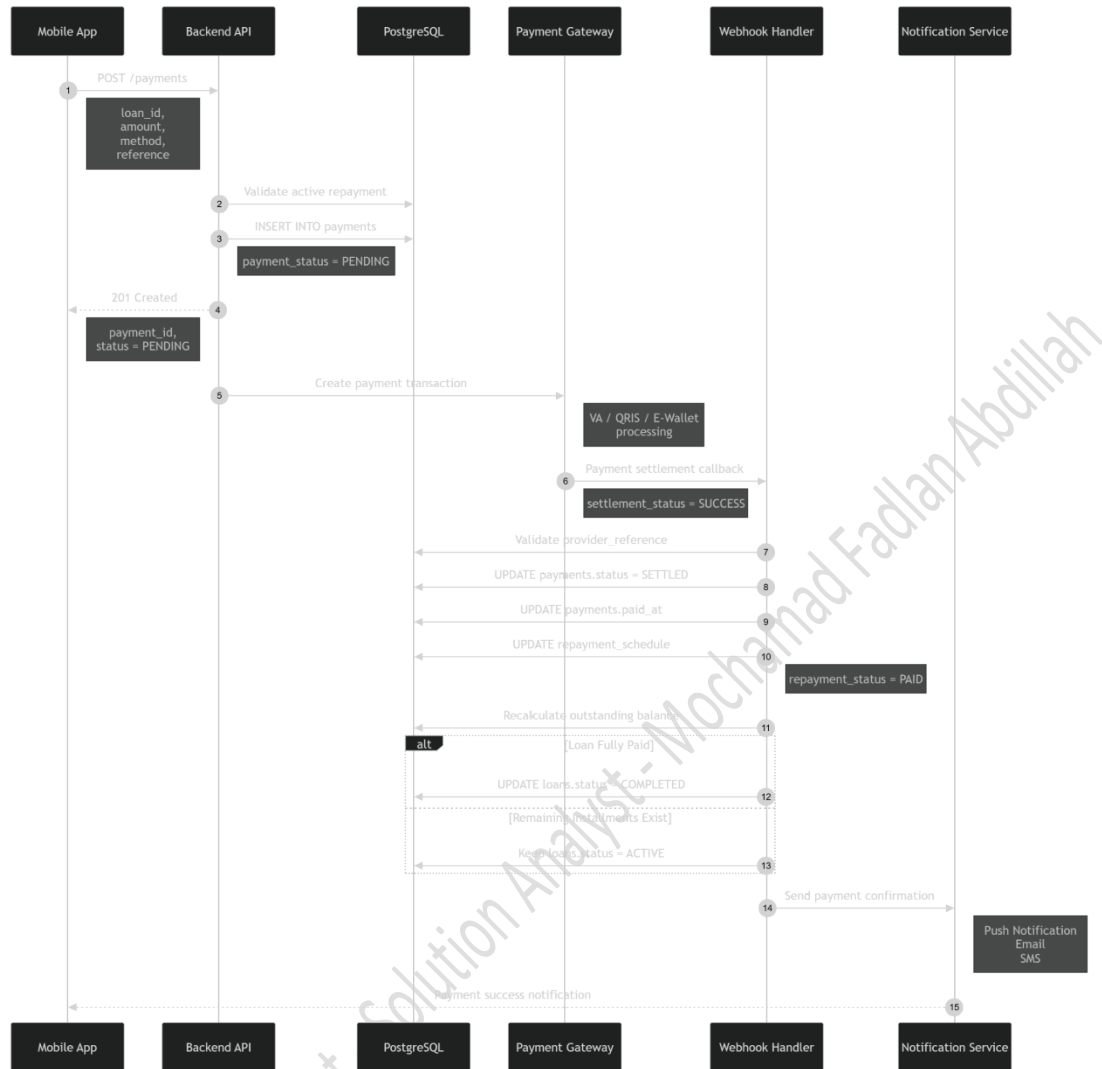
7.1 Registration & KYC Upload



7.2 Loan Application & Decision



7.3 Payment Settlement Cycle



8. Acceptance Test Matrix

Kode TC	Kasus Uji	Prasyarat	Langkah	Hasil yang Diharapkan
TC-01	Registrasi + Upload KTP Sukses	-	Register -> Upload KTP+Selfie -> Submit	User dibuat, kyc_status=verified
TC-02	Login Sukses + Biometrik Aktif	User exists	Login -> Enable Biometric -> close app -> buka pakai face ID/fingerprint	Token di-refresh, Dasbor tampil
TC-03	Ajuan Pinjaman Normal (Approved)	KYC verified, tidak ada pinjaman aktif	POST /loans 5jt/6bln	Loan pending -> approved -> jadwal cicilan dibuat
TC-04	Ajuan Pinjaman Ditolak (Conflict)	KYC verified, ada pinjaman aktif	POST /loans 5jt/6bln	Return 409 CONFLICT_ACTIVE_LOAN
TC-05	Ajuan Pinjaman Ditolak (KYC)	kyc_status=pending	POST /loans 5jt/6bln	Return 422 KYC_NOT_VERIFIED
TC-06	Tampilkan	Ada pinjaman aktif	Buka Dashboard	Outstanding balance

	<i>Outstanding Balance</i>			<i>sesuai</i>
<i>TC-07</i>	<i>Simulasi Webhook Settlement</i>	<i>Pinjaman aktif + jadwal cicilan ada</i>	<i>Bayar via VA -> Webhook settle -> Update</i>	<i>Payment settled, schedule updated</i>
<i>TC-08</i>	<i>Logout Bersih</i>	<i>User sudah login</i>	<i>Tap Logout -> kembali ke login screen</i>	<i>Token terhapus, biometric disabled</i>

9. MVP Implementation Plan

9.1 Sprint Breakdown

Sprint	Paket Pekerjaan	Estimasi
<i>Sprint 1</i>	<i>Backend skeleton: register, login, migrasi User + IdentityDocument</i>	<i>3-4 hari</i>
<i>Sprint 1</i>	<i>Signed upload URL, MinIO, /user/kyc endpoint, worker simulasi auto-verify</i>	<i>3-4 hari</i>
<i>Sprint 2</i>	<i>Model Loan, endpoint apply, worker decision engine scoring rule</i>	<i>4-5 hari</i>
<i>Sprint 2</i>	<i>Endpoint list loans, detail loan, repayment schedule</i>	<i>2-3 hari</i>
<i>Sprint 3</i>	<i>Endpoint payment, webhook handler, worker notifikasi</i>	<i>3-4 hari</i>
<i>Sprint 3</i>	<i>Implementasi layar mobile: daftar, login, KYC upload, dashboard, apply, detail, payment</i>	<i>5-7 hari</i>
<i>Sprint 4</i>	<i>Integration test, acceptance vectors, polishing, dokumentasi final</i>	<i>3-4 hari</i>

9.2 Total Estimate

Tim kecil (1-2 orang): estimasi 3-5 minggu untuk MVP end-to-end berjalan di environment test.

Single developer (1 orang): estimasi 5-6 minggu.

10. Error Code Reference

Kode HTTP	Kode Error	Deskripsi	Saran Penanganan Client
<i>400</i>	<i>INVALID_INPUT</i>	<i>Validasi parameter gagal</i>	<i>Periksa format payload request</i>
<i>401</i>	<i>UNAUTHORIZED</i>	<i>Token akses/refresh tidak valid atau kedaluwarsa</i>	<i>Refresh token atau login ulang</i>
<i>403</i>	<i>FORBIDDEN</i>	<i>Akses resource ditolak</i>	<i>Periksa hak akses user</i>
<i>409</i>	<i>CONFLICT_ACTIVE_LOAN</i>	<i>Ada pinjaman masih</i>	<i>Lunasi kewajiban</i>

		<i>berjalan (aktif/pending)</i>	<i>sebelum ajukan baru</i>
422	KYC_NOT_VERIFIED	KYC belum diverifikasi	Selesaikan upload KTP dan selfie
422	INSUFFICIENT_DATA	Data diri kurang lengkap	Lengkapi data yang kosong
429	RATE_LIMITED	Terlalu banyak percobaan login gagal	Tunggu beberapa menit sebelum coba lagi
500	INTERNAL_ERROR	Terjadi error internal server	Hubungi customer support PT. XYZ

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